

Ahmed Ahmed

Ahmedsqrdd@gmail.com • +61 479 088 469 • www.linkedin.com/in/ahmed-mohamoud-ahmed

OBJECTIVE

Penultimate-year Mechanical & Mechatronics Engineering student at the University of Technology Sydney with current industry experience at a Sydney engineering consultancy. Strongest where engineering analysis meets software – structural fatigue, FEA and thermofluids alongside Python, C++ and real-time control implementation. Experienced leading a multidisciplinary hardware project from schematic to validated prototype.

EDUCATION

Bachelor of Engineering (Honours) - (Mechanical & Mechatronics)

University of Technology Sydney

Credit Average (WAM - 70.1)

2023 | Expected
2027

Key results: Applied Mechanics & Design B- High Distinction, Mathematics 2 – High Distinction, Designing Sustainable Engineering Projects – Distinction, Sensors & Control for Mechatronic Systems – Distinction, Thermofluids A - Distinction

TECHNICAL SKILLS

Programming languages: Python, C, C++, JavaScript, MATLAB

CAD & Analysis: AutoCAD, SolidWorks, Bluebeam, Revit, FEA.

Tools & Frameworks: OpenCV, ROS, Altium Designer, Next.js, Supabase, Git & GitHub.

Engineering: Structural & fatigue analysis, thermofluids, PCB design & validation, embedded firmware, computer vision, closed-loop control, hydraulic design

Soft Skills: Technical Communication, Leadership, Process optimisation and Stakeholder Engagement.

PROJECTS & EXTRACURRICULAR

Robotic Visual Servoing System — Sensors and Control for Mechatronic Systems | D

2025

- Implemented an image-based visual servoing (IBVS) system enabling a Universal Robots UR3 manipulator to track a target in real time using an eye-in-hand Intel RealSense D435 RGB-D camera.
- Built the vision pipeline in Python with OpenCV and pyrealsense2: checkerboard detection with sub-pixel corner refinement, median-filtered depth estimation and image-space error computation feeding the interaction matrix.
- Implemented the IBVS control law, mapping camera-frame velocity to joint velocities through the Jacobian pseudo-inverse and driving the robot over a ROS bridge in a closed loop at 15–20 Hz.
- Diagnosed latency in the initial MATLAB and WSL architecture and migrated the system to native Python, then tuned control gain, velocity smoothing and joint limits to achieve stable convergence without oscillation.

- Led a four-person team building a custom PCB step tracker (Arduino Uno R3, ADXL335 MEMS accelerometer, OLED interface), coordinating mechanical, electronics and firmware workstreams against a fixed delivery timeline.
- Designed the power regulation subcircuit end to end in Altium Designer – reverse-polarity protection, 5 V and 3.3 V LDO rails, decoupling and thermal dissipation calculations – validating both rails within $\pm 5\%$ tolerance under a dynamic load.
- Authored the step detection firmware in C++, implementing startup calibration, hysteresis thresholding on acceleration magnitude to reject false triggers, and circular-buffer cadence computation; achieved 94% step-count accuracy over a 100-step walking test.
- Carried the board from schematic through layout to fabrication and bench testing, passing design rule verification with zero violations.

Engineering Onboarding & Competency-Tracking Web Application – Self-directed**2026 – Present**

- Designed and built a modular web application that structures engineering onboarding into staged competency tracks with objectives, checklists and curated technical references.
- Migrating the platform to Next.js and Supabase to add authentication and a query able progress database, so time-to-competency can be measured directly rather than estimated.

PROFESSIONAL EXPERIENCE

Undergraduate Hydraulic Engineer | Goldfish & Bay**Feb 2026 – Present**

- Prepare hydraulic services designs and documentation across concept, tender and construction phases under the direction of senior engineers.
- Undertake hydraulic calculations, drawings, markups and design revisions to a review-ready standard, working to Australian Standards and compliance requirements.
- Attend site inspections of new and existing buildings, assessing installed hydraulic systems against design intent and compliance obligations.
- Identified a slow, error-prone manual sizing process and independently built an internal schematic sizing tool now used by the team.
- Proposed and leading the development of the firm's online onboarding learning modules, producing structured technical content for incoming engineers.

Sales Consultant & Samsung Elite Representative | Optus**Jul 2024 – May 2026**

- Delivered technical product demonstrations following Samsung Elite certification, translating detailed specifications into clear customer benefits.
- Mentored team members on product knowledge and troubleshooting, reducing ramp-up time for new starters.
- Consistently exceeded monthly sales targets and maintained CRM data used for lead tracking and pipeline forecasting.

Business Development Consultant | Chandler Macleod**Oct 2023 – May 2024**

- Conducted market research and outbound business development, qualifying and pitching tailored recruitment solutions to decision-makers nationwide.
- Coordinated with recruiters nationally to communicate client requirements and ensure a smooth handover of sales to service delivery.
- Utilised CRM software to document client interactions, track activities and generate insightful reports for management.